

Design and Demonstration of an OFDM Based RadCom System

Grant Norrie, Stephen Paine*
Department of Electrical Engineering
University of Cape Town
Cape Town, South Africa
*stephen.paine@uct.ac.za

Abstract—A joint Radar Communications testbed is presented. This testbed leverages the OFDM based DAB standard to generate Radcom signals. The extended DAB mode structure used to describe these signals was used as the basis on which the communications sub-systems were designed. Furthermore, a radar processing subsystem was developed to process the same signal. Finally a functional testbed was deployed and used to complete system integration tests thereby demonstrating the joint RadCom functionality.

Index Terms—DAB, OFDM, EM, IOO

I. INTRODUCTION

Over the past decades the price of electronic hardware has fallen while performance has continually improved. This increase in performance and corresponding increase in accessibility has led to the broader use and integration of the electromagnetic spectrum usage and RF hardware into modern electronics. This broader use is not restricted to radar applications but is shared with increasingly prevalent communication applications. This growing usage of the EM spectrum usage combined with surrounding regulations has led to the spectrum becoming increasingly congested.

This has prompted the development of novel methods for more efficient spectrum usage. Of particular interest is the growing usage of Orthogonal Frequency Division Multiplexing (OFDM) signals in communications standards. As an example: FM radio transmissions are being replaced by the OFDM based Digital Audio Broadcasting standard [1]. Alongside these communications developments is the developing field of passive radar which aims to take advantage of illuminators of opportunity (IOO). Transmitters not intended for radar applications rather transmitting for their own purposes such as communications.

An example of work in this field is a previously developed Digital Audio Broadcasting (DAB) based passive radar processing chain [2]. Such a system has its traditional radar applications but raises a question. If a passive radar system is able to use a communication IOO's for radar applications, then could a system itself not use its own communication transmissions for radar applications? Or conversely, could a system use its own radar transmissions for communication purposes? This type of system will be termed a RadCom system.

A. Literature Review

This implementation, termed co-designed dual function systems by [3] uses a single waveform to achieve both communications and radar functionality. Alongside such a system are co-existing radar communications systems. These are systems that may interleave communication and radar waveforms in time and space [3].

Co-designed dual function systems are the focus of this paper, therefore notable investigations surrounding OFDM based RadCom system are deemed important to review. [4] provides an overview of an OFDM based RadCom system. This includes the analysis of communications and radar processing for different system topologies. [4] additionally discusses multi-user scenarios going on to note the challenges an OFDM based RadCom system may face due to the high peak-to-average power ratio. [5] completed an investigation of this topic proposing methods for reducing peak-to-average power ratio of OFDM based RadCom waveforms.

Outside of investigations surrounding waveforms, system topologies and general processing techniques, is the investigation of the radar performance of such a waveform. [6] completed this work surrounding the estimation of range and Doppler processing using RadCom waveforms.

II. THE DAB STANDARD

In light of the research of OFDM based RadCom systems, the OFDM based DAB standard was used as the foundation of this investigations RadCom system development. It is fully defined in the ETSI DAB standard [1]. This standard implements $\frac{\pi}{4}$ -DQPSK to encode audio waveforms onto OFDM signals. The DAB signal is much like a standard OFDM signal, comprising of multiple sub-carriers contiguous in frequency with an exception. In the case of the DAB standard the central carrier is nulled [1]. Multiple contiguous symbols form what is termed a frame and multiple contiguous frames form the complete DAB signal [1].

Within the scope of system development further specifics surrounding a DAB signal are simplified and split into three components. These include a Null Symbol (NS), Phase Reference Symbol (PRS) and what are termed data carrying symbols. The Null Symbol is the first symbol in a DAB frame with all sub-carriers weights set to zero while the Phase Reference Symbol is the second symbol transmitted and is

fully defined in the DAB standard [1]. The PRS is therefore known *a priori*, defined by complex phase weights applied to the 1536 active carriers used by the DAB standard. This is used for frame detection and fine time synchronisation. This provides a DAB system with the ability to demodulate and decode signals. The DAB standard abstracts the OFDM signal description into a single structure referred to as the DAB mode [1].

III. SYSTEM OVERVIEW

The DAB based RadCom system is comprised of four sub-systems. These are defined as the Signal Generation Chain, Hardware Testbed, Radar Processing Chain and Communications Processing Chain. An overview of their relationships is shown in the Figure 1. Combined, these systems allow for the design, customization and deployment of a RadCom system for demonstration purposes.

The Signal Generation Chain is responsible for the encoding of binary bit-streams onto OFDM waveforms specifically designed for both radar and communication applications. The Hardware Testbed is responsible for the transmission and receipt of the waveforms previously generated. A received waveform is recorded and processed by both the Communications Processing Chain and Radar Processing Chain. This resulted in the extraction of the initially encoded bit-stream and the generation of range-Doppler plots. This RadCom system, using a single test setup and the described subsystems, was able to complete both radar and communications processing on a single received waveform.

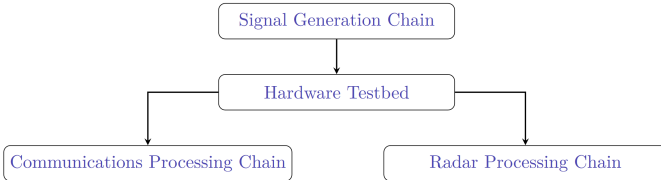


Fig. 1. Showing the RadCom Subsystems and their relationships

A. Signal Generation

The Signal Generation Chain generates a DAB styled OFDM signal. These signals differ from those defined in the DAB standard in two ways. Additional parameters were introduced into the signal definition while the exact parameters used to generate the signals were additionally altered to better suit the RadCom application.

This signals fundamental frequency domain component is the single sub-carrier k which is defined for a symbol period T_s . This symbol period is defined as a fixed number of samples where the sample period T is the inverse of the processing chains sampling rate f_s . Furthermore the symbol period is further defined as the summation of the guard and integration period, T_g and T_u , where the guard interval is a cyclic prefix of the integration period.

K sub-carriers spaced at $\frac{f_s}{T_u}$ Hz form a single symbol l . L consecutive symbols in time form a single frame m while M frames form the complete active portion of the signal. Each sub-carrier in the signal is scaled by a real weight and complex phase coefficient defined according the sub-carrier, symbol and frame indices. These are denoted using $\omega_{m,l,k}$ and $a_{m,l,k}$. This active portion of the signal is described in Equation 1.

$$s(t) = \sum_{m=-\infty}^{\infty} \sum_{l=0}^L \sum_{k=-\frac{K}{2}}^{\frac{K}{2}} (\omega_{m,l,k}) (a_{m,l,k}) \exp\left(\frac{j2\pi k(t - T_g)}{T_s}\right) \text{rect}\left(\frac{t}{T_s}\right) \quad (1)$$

The Signal Generation Chain additionally provides the ability to append delay times thereby providing the ability to toggle between continuous and pulsed modes of operation. This delay time T_d is described as a fixed number of processing chain sample periods.

Signal parameters that remain unchanging across frames are abstracted into an expanded DAB mode structure to facilitate processing across RadCom sub-systems and is shown in Table 1. These include additionally configurable parameters including hardware sampling rates and transmission carrier frequencies.

Parameter	Description
K	The number of OFDM carriers in a single symbol
L	The number of symbols (including NS and PRS) in a frame
T_{null}	The length of the null symbol in elementary units
T_u	The length of the integration period in elementary units
T_g	The length of the guard interval in elementary units
T_s	The length of a single symbol in elementary units
T_d	The length of a delay period in elementary units
f_h	The sampling rate used by the hardware during TX/RX
f_s	The processing chain sampling rate
T	The inverse of the processing chain sampling rate
f_c	The carrier frequency of the RadCom waveform

TABLE I

SHOWING THE EXPANDED DAB MODE STRUCTURE INCLUDING ALL PARAMETERS RELEVANT TO THE RADCOM SYSTEM

This DAB mode structure is used to generate a signal in three distinct phases. First, the bit-stream which one intends to transmit, is converted to the complex phase weights corresponding to one of the following phase values $[0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}]$. Importantly, the phase weights corresponding to $l = 1, 2$ are the weights for the NS and PRS as defined in the DAB standard. All subsequent symbols phase weights are computed using $\frac{\pi}{4}$ -DQPSK with the PRS ($l = 1$) as the reference. The output of this procedure corresponds to the complex phase weights $a_{m,l,k}$. This procedure is denoted as Bit Encoding and is shown in Figure 2.

This is followed by frame generation where M frames, each containing L symbols, which in turn each contain K sub-carrier. Each of these sub-carriers is scaled by the complex phase coefficient $a_{m,l,k}$ (previously generated) and unity magnitude sub-carrier weight coefficient corresponding to $\omega_{m,l,k}$. Finally, the data carrying portion of the signal is appended with the delay period corresponding to time insertion in Figure 2.

An important nuance to signal generation was the method used to generate an adjusted PRS. The DAB standard defined complex phase weights of the PRS as a function of sub-carrier index with respect to the DAB central null carrier [1]. Therefore to leverage this fact, the Signal Generation Chain extracts the central K phase weights and applies them to the RadCom sub-carriers.

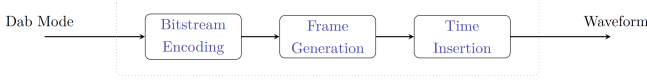


Fig. 2. Showing a block diagram of RadCom signal generation

1) *Communication Data Rates:* The DAB mode structure is also a useful abstraction to generalise signal processing and provides a method to quantify the possible communication data rates in bits per second (bps). To begin, the number of bits encoded in a single differential phase is 2 derived from base 2 logarithm of the 4 phases used to generate the waveform. Each of these bits are mapped to the K carriers present in a single symbol. Additionally, L symbols in a given frame. Recall, as per the DAB standard, the first two symbols are not considered ‘data carrying’. This defines the number of bits encoded in a single waveform and shown in the numerator of the fraction below.

$$bps = \frac{2 K (L - 2)}{T_{null} + (T_s (L - 1) + T_d)} \quad (2)$$

Following this, the time it takes to transmit these bits shall be determined. Firstly, recall that the symbol time is comprised of the integration and guard interval: $T_s = T_g + T_u$. One should also recall that there are L symbols in a given frame. Notably, only $L - 1$ are of length T_s as the first symbol, the NS is of length T_{null} . The final addition to the frame time is the inclusion of the delay time T_d . This is the denominator of the equation above and combined will define the bit rate of the designed waveform.

B. Communication Processing Chain

The Communications Processing Chain was developed as an extension to a previously developed DAB processing chain [2]. This included the extension of a preprocessing and demodulation functionality and the addition of decoding functionality as shown in Figure 3. Preprocessing, broadly defined, detects the and extracts a single frame. Demodulation then converts the frame to the modulated phase weights while decoding converts phase weights to the original bit-stream.

Preprocessing corresponds to fine time synchronisation and therefore the customized PRS. The communication chain therefore generates it own PRS using the same customised DAB mode as that used to generate the received signal. This PRS is used to create a matched filter which is applied as the processing chain increments through a recording. If this matched filtering procedure returns a max value above a threshold defined at the time of operation, a frame is

considered to be present. The remaining frame after the null symbol is then extracted and passed on for demodulation.



Fig. 3. Showing a block diagram of RadCom signal communication processing

During demodulation the frame is split into its consecutive symbols. Once split the guard intervals are removed, leaving only the integration periods worth of samples in each symbol. An FFT of these symbols is taken followed by the consecutive symbol wise complex division of each sub-carrier as shown in Equation 2. This procedure is the mathematical inverse of the bit-stream encoding implemented during Bit Encoding. At this point $L - 2$ data carrying symbols remain after the removal of the null symbol and usage of the PRS as the reference symbol. The phases of this complex division process are then extracted and passed to the decoding procedure.

$$D_{l-1} = \angle \left(\frac{L_l}{L_{l-1}} \right) \quad (3)$$

These phases are inverse mapped from their phase value to their equivalent bit set and concatenated into a continuous bit-stream. At this point the extracted bit-stream is identical to that originally encoded when generating the symbol using the Signal Generation Chain.

C. Radar Processing Chain

The radar processing chain computes a range-Doppler map using passive radar techniques through cross correlating of a reference and surveillance signal. The processing chain therefore encompassed two procedures. The first covered time alignment, aligning the surveillance and reference channel. This alignment procedure, similar to that used in the Communications Processing Chain, involves the detection and alignment of the first PRS in the reference and surveillance signal.

The second is the computation of the range-Doppler map. Once aligned, the processing chain uses the DAB mode structure to slice both the surveillance and reference channel into chunks corresponding to the pulse duration (or the slow time slice). This is the combination of the active signal time and the additional delay period. These slow time slices were therefore non overlapping and contiguous in time. This results in the effective processing chain being the inverse of the PRT where the PRT is equal to $L(T_g + T_u) + T_d$. The cross-correlation of the reference and surveillance signal chunks are computed, followed by the FFT computation along the slow time dimension. This results in the generation of the range-Doppler map.

Finally, the radar processing chain additionally facilitated additional processing techniques such as clutter map removal and pulse cancellation. pulse cancellation is the procedure described by [4] which involved the subtraction of slow time

slice $N - 1$ from slow time slice N . These procedures remove static clutter from range-Doppler maps.

1) *Range Resolution*: Beyond the processing techniques implemented in the radar processing chain is the radar performance of the RadCom waveform. This performance may be described in terms of the DAB mode structure.

One may describe the range resolution of such a waveform through the use of $\Delta R = \frac{c}{2BW}$. This can be done by multiplying the number of carriers, K , by their frequency separation Δf . Recall that the frequency separation is defined by the inverse of the integration time T_u . Therefore the range resolution of a signal defined by the DAB mode can be defined as follows:

$$\Delta R = \frac{c}{2K T_u} \quad (4)$$

2) *Doppler Resolution*: The Doppler resolution may additionally be described as the inverse of the coherent integration time. This is the inverse of the frame time as is stated below.

$$\Delta V = \frac{1}{T_{null} + (T_s(L - 1) + T_d)} \quad (5)$$

D. Hardware Testbed

The testbed was fundamental to the demonstration of the RadCom sub-systems but faced inherent complications. These complications are due to isolation limitations of the USRP N210 which resulted in the selected testbed topology as shown in Figure 4. This configuration uses two USRP 210s Ettus A and B which are designated as the transmitter and receiver, respectively. Two Ettus' were used to increase the distance and therefore isolation between transmit and receive ports as demonstrated by [3]. The two devices were synchronised using a MIMO cable as done by [3].

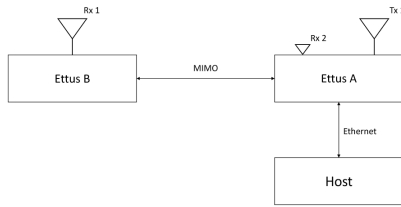


Fig. 4. Showing the hardware testbed configuration used for RadCom system integration tests

Both the transmitter and receiver ports were attached to 2.45 GHz can antennas while a third port, a receiver on Ettus A, was terminated with a 50Ω as shown in Figure 4. The MIMO cable enabled communication and synchronisation between the two devices while an ethernet cable connected Ettus A to a host PC. This host connection allowed for the streaming of data between host and the Ettus' and additionally facilitated the transmission of start and stop commands.

Rx 2 was included in this testbed and used during radar processing as it provided the testbed the ability emulate a passive radar system. This approach was taken primarily due to hardware availability constraints.

IV. SYSTEM INTEGRATION TESTS

The system integration tests were designed to demonstrate the dual radar and communications functionality. This included a direct link communications test, a static range test and a moving target test. All tests used the testbed configuration shown in Figure 4 with the set of hardware transmission parameters that remain similar across tests shown in Table 2.

Parameter	Symbol	Value
Tx Gain	-	0 dB
Rx Gain	-	0 dB
Tx Bandwidth	-	1.1 MHz
Rx Bandwidth	-	1.1 MHz
Sampling Rate	f_h	5 MHz
Carrier Frequency	f_c	2.45 GHz
Read Type	-	Float
Write Type	-	Double

TABLE II

SHOWING SDR PLATFORM PARAMETERS USED DURING THE DIRECT LINK COMMUNICATIONS TEST

A subset of the DAB mode transmission parameters are shown in Table 3. The waveform is designed for no other purpose than to test RadCom functionality, therefore in reality values used in a deployable system may vary. Notably, for the purposes of testing the NS was removed due to the inclusion of the delay period.

Two symbols, each containing 500 sub-carriers, were used during the System Integration Tests. Symbol 1 was therefore the adjusted PRS while symbol 2 was a data carrying symbol. These two symbols had an integration and guard period of 2048 and 504 samples respectively. Furthermore the delay period was 5104 samples while processing sampling rate was 4.96 MHz

Parameter	Value
f_s	4.96 MHz
K	500
L	2
T_u	2048
T_g	504
T_{null}	0
T_d	5104

TABLE III

SHOWING DAB MODE CONFIGURATION USED DURING THE DIRECT LINK COMMUNICATIONS TEST

A. Communications Demonstration

The communication test aimed to verify the communications sub-system functionality of the RadCom system but not map performance across varying scenarios. This test was therefore conducted by separating the TX and RX antennas and first testing transmission levels until no feed through was detected on the RX port. Once one was able to ensure that demodulated bits are not a result of direct feed-through the communications test could be conducted.

With the absence of feed through verified the TX and RX antennas were separated by 5 meters and directed towards one another. The RadCom signal was transmitted towards the

RX antenna with the received signal recorded and stored. Following this, the signal was passed through the communications processing and chain demodulated. The BER rates were calculated through the comparison of the initially encoded bit-stream and the decoded bit-stream. The BER result was the expected 0% indicating that the communications chain works correctly under high SNR conditions.

B. Static Range Test

The static range test used a Hyundai i20 with a trihedral corner reflector placed on the dashboard as the target. The test used the same waveform parameters and similar system parameters as that used in the communications demonstration. The difference between the two tests being the transmit and receive gain. These were set to 30 and 20 dB. The same two can antennas were used as the transmitter and receiver. One antenna was fed with a horizontally polarized waveform while the second with a vertically polarized waveform to increase their isolation. RX 2 was terminated with a 50 Ω resistor and sampled, measuring the feed-through as an emulated reference channel to facilitate additional processing.

The direct feed through due to the dual RX/TX configuration introduced the need for two measurements. The first was a reference clutter map measurement, a range-Doppler image of the environment without a target present. The second was the range measurement with the target positioned 100 m away from the testbed. The two recorded signals were stored and passed through the radar processing chain. When processing the recorded data, RX 1 was treated as the surveillance channel while RX 2 was considered the reference channel.

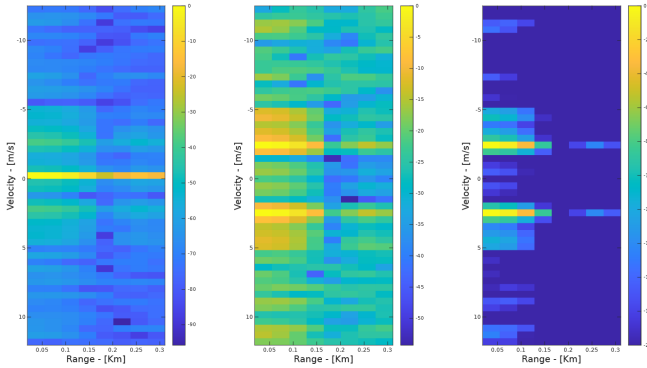


Fig. 5. Showing static range test range-Doppler maps including a standard range-Doppler map, pulse cancelled range-Doppler map and pulse cancelled adjusted dynamic range-Doppler map

The above plots from left to right show an unadjusted range-Doppler map, a range-Doppler map with pulse cancellation applied and finally a range Doppler map with pulse cancellation applied coupled with and adjusted dynamic range. This response of the imaged environment with the target present at 100 m demonstrates that the target is not identifiable without further processing. This inability to detect the target is primarily due to direct feed through between the TX and RX antennas.

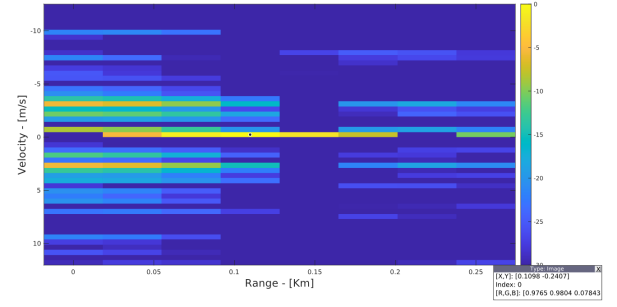


Fig. 6. Static range test with clutter map removal applied demonstrating the presence of a target at 100 m

This warranted the inclusion of an additional clutter map removal processing step. The result of this is shown in Figure 6. Instead of removing stationary targets as done by pulse cancellation, this methodology removes the clutter power, as initially imaged, from the range-Doppler map. With this power removed one would expect the target to appear at 100 m. This is what occurs with a target response occurring in the expected 109 m range bin. The response was expected in this range bin as this is the closest range match with the actual target range of 100 m for the given system parameters.

C. Range-Doppler Test

The same hardware and software configuration used in the static range test was used to complete the combined range-Doppler test. The test involved three measurement mechanisms. The first was the odometer reading in the car, the second used a derived velocity calculation through the use of GPS measurements and the third was the velocity measurement using the RadCom testbed. The range-Doppler test involved a target positioned 30 m in front of the hardware and moved away from the hardware at a planned speed of 25 kmh^{-1} .

Parameter	Value
GPS Average Velocity (ms^{-1})	7.2
GPS Average Velocity (kmh^{-1})	25.9
GPS measurements	10
Odometer Velocity (kmh^{-1})	25
Odometer Velocity (ms^{-1})	6.9
Range Bin (m)	30
True Range (m)	36

TABLE IV
SHOWING THE TRUE AND RECORDED PARAMETERS OF THE RANGE-DOPPLER TEST WITH A TARGET AT A RANGE OF 30 M AND MEASURED VELOCITY 7.4 MS^{-1}

The results of this test are shown in Figure 7. The first subplot shows the range-Doppler map without additional processing. This demonstrates that the target is not discernible due to the feedthrough between TX and RX. The second and third subplots have pulse cancellation applied. The dynamic range on the third subplot was also adjusted to highlight the presence of the target. With pulse cancellation applied and the static clutter removed, a target is clearly visible.

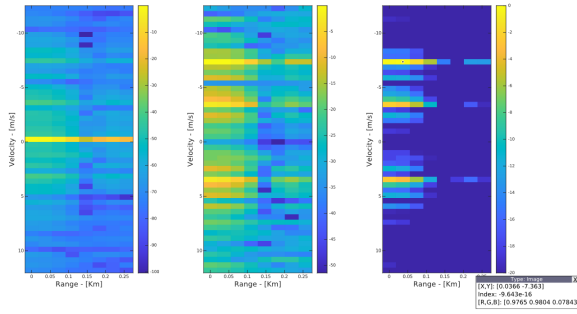


Fig. 7. Showing range-Doppler maps of a target moving away from the testbed at a speed of 7.4 ms^{-1}

The measured results of this test are shown in Table 4 with the targets true range being 30m while the GPS and odometer velocities were 7.2 and 7.0 ms^{-1} respectively. This is compared to the measured range and velocity of 36 m and -7.4 ms^{-1} respectively.

V. CONCLUSION

This investigation was able to expand the DAB standards physical signal for use in a RadCom System Integration Test. This was achieved using an adjusted DAB mode structure which provides a method to design this signal through the alteration of its time and frequency domain parameters. This therefore provides the RadCom system with customizable radar and communications performance achieving both radar and communications functionality.

The signal generation chain, using the RadCom DAB mode provides two specific functionalities in the communications chain. The first is the ability to generate the RadCom OFDM signal while the second introduces the ability to encode bit-streams onto OFDM waveforms using $\frac{\pi}{4}$ -DQPSK.

This combined with the refactored DAB passive radar processing chain and the radar processing chains ability to generate range-Doppler maps using the DAB mode to derive signal information provided the software functionality to demonstrate a functional RadCom system.

Furthermore, it was demonstrated that using the Ettus N210s it was possible to create a testbed that was both able to simultaneously transmit and receive signals. This allowed for the real world independent demonstration of the communications and radar functionality using the same waveform. This was demonstrated through the calculation of the BER during the communications tests and the generation of range-Doppler maps with targets during the radar tests which contained both stationary and moving targets.

REFERENCES

- [1] Radio Broadcasting Systems; Digital Audio Broadcasting (DAB) to mobile, portable and fixed receivers, ETSI EN 300 401, ETSI, France, 2006
- [2] C. Tilbury, "DAB Processing Chain for Passive Radar Applications" B.S. Thesis, Dept Electrical Eng, UCT, Cape Town, 2020.
- [3] F. Liu, C. Masouros, A. Li, H. Sun and L. Hanzo, "MU-MIMO Communications With MIMO Radar: From Co-Existence to Joint Transmission," in IEEE Transactions on Wireless Communications, vol. 17, no. 4, pp. 2755-2770, April 2018, doi: 10.1109/TWC.2018.2803045.

- [4] L. Giroto de Oliveira, B. Nuss, M. B. Alabd, A. Diewald, M. Pauli and T. Zwick, "Joint Radar-Communication Systems: Modulation Schemes and System Design," in IEEE Transactions on Microwave Theory and Techniques, vol. 70, no. 3, pp. 1521-1551, March 2022, doi: 10.1109/TMTT.2021.3126887.
- [5] Y. Huang, S. Hu, S. Ma, Z. Liu and M. Xiao, "Designing Low-PAPR Waveform for OFDM-Based RadCom Systems," in IEEE Transactions on Wireless Communications, vol. 21, no. 9, pp. 6979-6993, Sept. 2022, doi: 10.1109/TWC.2022.3153606.
- [6] J. -F. Gu, J. Moghaddasi and K. Wu, "Delay and Doppler shift estimation for OFDM-based radar-radio (RadCom) system," 2015 IEEE International Wireless Symposium (IWS 2015), Shenzhen, China, 2015, pp. 1-4, doi: 10.1109/IEEE-IWS.2015.7164629.
- [7] M. Fuhr, M. Braun, C. Sturmz, L. Reichardt, F.K. Jondral, March. 2012 "An SDR-based experimental setup for OFDM-based radar". In Proceedings of the 7th Karlsruhe Workshop on Software Radio (p. 176)
- [8] F. Le Chevalier, O. Krasnov, F. Deudon and S. Bidon, "Clutter suppression for moving targets detection with wideband radar," 2011 19th European Signal Processing Conference, Barcelona, Spain, 2011, pp. 427-430.